Date :
Page :

Experiment - 13

Aim: To study and compare SRAM, EEPROM and Flash memory based on persistence, speed and Endurance.

Apparatus Required:
- Computer system / Microcontroller kit
- Datasheets of memory devices
- Internet / Reference material.

Theory:
Memory devices are used to store data in digital systems. They are broadly classified into volatile and non-voltatile memory
- Volatile Memory losses data when power is OFF
- Non-volatile Memory retains data even without power

1) SRAM (Static Random Access Memory)
SRAM is a type of volatile memory that stores data using flip-flops.
Features:
Very fast access time
Does not require refreshing
Expensive and low storage density
Data is lost when power is OFF

2) EEPROM (Electrically ~~Era~~ Erasable Programmable ROM)
Definition
EEPROM is non-volatile memory that can be electrically erased and reprogrammed
Features:
- Data retained without power
- Can be erased byte-by-byte
- Slower than SRAM

AMAR KRISH

Date :
Page :

· Limited write cycles

3) Flash Memory

Flash memory is a type of EEPROM that erases data in block instead of bytes.

Features:

· Non-volatile
· Faster than EEPROM
· used in USB drive, SSDs
· erased in blocks.

Results

The characteristics of SRAM, EEPROM, and flash memory were studied and compared succesfully based on persistance, speed and endurance.

AMAR KRISH

Teacher's Signature